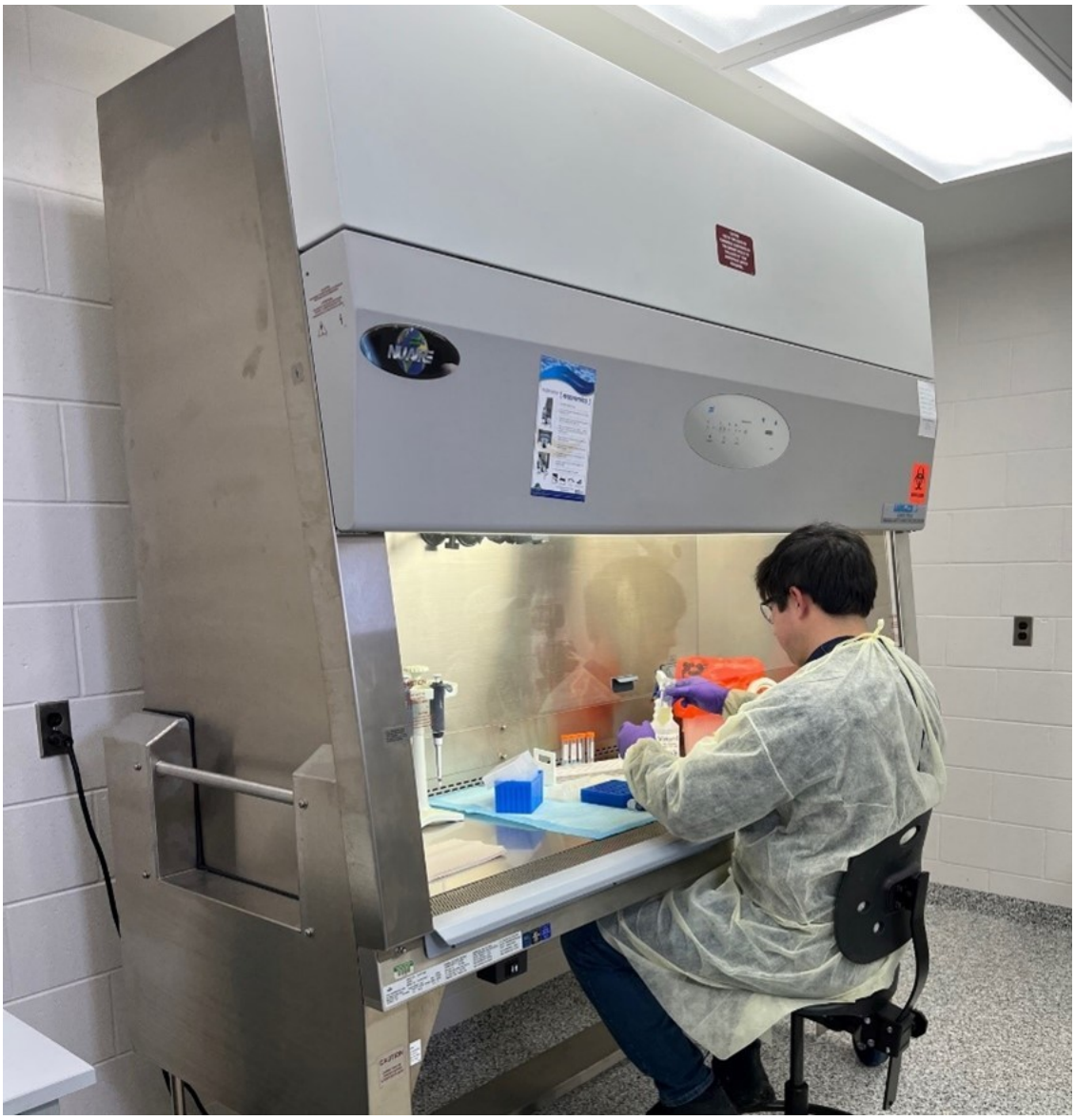
Tab 1

A drug testing laboratory needs a new database to track the various aspects of its testing services.

- Employees
    - The lab keeps track of employee ID, position title, name, years with the company, and yearly salary
    - Employees can perform many tests some days but may only perform one test on other days
- The laboratory performs different tests for its clients. The system should track information about the tests that are performed.
    - This includes the test ID, the employee who performed the test, the materials used to perform the test, and the client that the test is being done for
- The laboratory wants to keep track of its materials in stock by recording the lot number and expiration date of each material
- The laboratory has different fume hoods. They want to keep track of which hoods were used during which test in case of any contamination issues.
    - The date of the last inspection of the fume hood should be recorded as well as the result (pass/fail), and the technician who inspected it
    - The company wants to track which department the fume hoods are in
    - Each fume hood also has equipment in it (like pipets) that the system should track
- Technicians are outside contractors who inspect fume hoods in the lab. Each technician can inspect any of the fume hoods and can be responsible for checking multiple hoods.
- There are 3 different departments in the laboratory (sample processing, testing, analysis and reporting)
- The laboratory performs its testing services for different companies
    - The system should track the company name, the name of the person to contact at the company, the phone number, email, and mailing address



A fume hood

Project Proposal

Ryan Bennett
Chirag Chirag
Christina Kuleshevich
Anusuya Roysarkar
Emily Tellschow

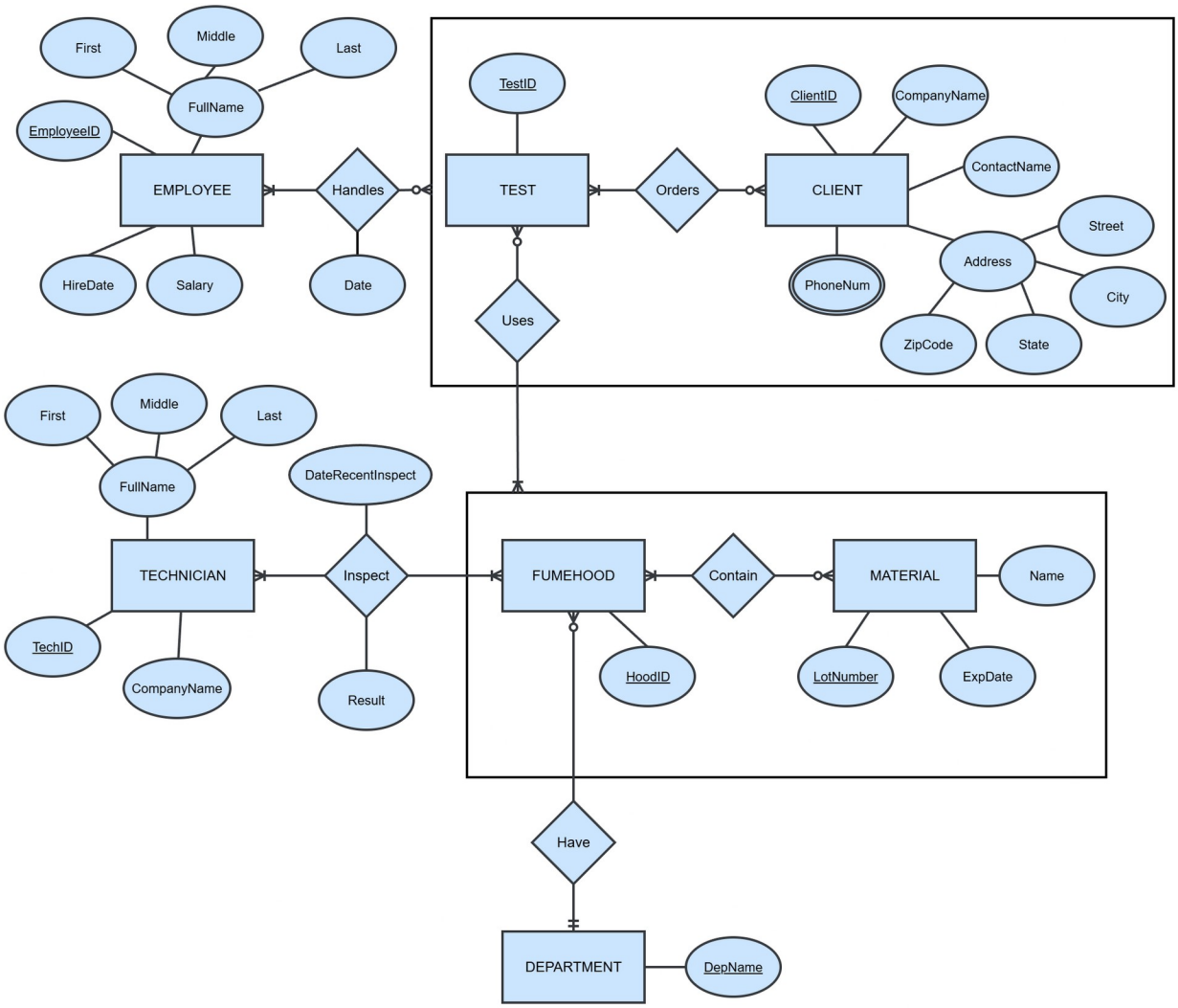
Project Proposal

For our project, we will be focusing on building out a database for a drug testing laboratory. The laboratory wants to track a variety of information related to the tests they perform, their clients and the equipment they use. They need a system that would store all the information of their clients, employees, tests, and materials. In general, work is initiated at the lab when clients order tests. These drug tests are then performed by employees in fume hoods using chemicals/materials and the results are sent back to the clients. While the lab has a separate system for organizing the results of the tests, they do want this database to include information about the clients and link that information to the tests that are being performed for them. It will also contain information about employees.

Due to the volatility and danger of some of the chemicals used in the tests, all tests are performed in fume hoods (see Picture 1 in Supplementary Materials.) These fume hoods have filters and fans within them that pull air from the outside into the hood through it, as well as a shash (a sliding glass panel at the front of the hood) that can be raised or lowered and to ensure no toxic fumes escape into the room (Picture 2). The fume hoods must be used to ensure that employees are not exposed to these chemicals and also have routine maintenance that the lab wants to track.

This database will allow the laboratory to organize and easily access information about their employees, tests, clients, and laboratory equipment as well as how those things relate to each other. It will allow the laboratory to determine which materials and fume hoods were used for specific tests through foreign keys and junction tables, which is important from a quality assurance standpoint should a test fail. The laboratory will also be able to compile and streamline existing information into one database that can be easily navigated and accessed by management.

ER Diagram



Relational Schema

-

1. EMPLOYEE (EmployeeID, First, Middle, Last, HireDate, Salary)
2. TEST (TestID)
3. CLIENT (ClientID, CompanyName, ContactName, Street, City, State, ZipCode)
   CLIENTPHONENUM (PhoneNum, ClientID[FK])
4. ORDERS (TestID[FK], ClientID[FK])
5. HANDLES (EmployeeID[FK], TestID[FK], ClientID[FK], Date)
6. FUMEHOOD (HoodID, DepName[FK])
7. DEPARTMENT (DepName)
8. MATERIAL (LotNumber, Name, ExpDate)
9. CONTAIN (HoodID[FK], LotNumber[FK])
10. USES (TestID[FK], HoodID[FK], LotNumber[FK])
11. TECHNICIAN (TechID, First, Middle, Last, CompanyName)
12. INSPECT(TechID[FK], HoodID[FK], DateRecentInspect, Result)

New data Variant 2 (tables)
Laboratory Variant 2.xlsx

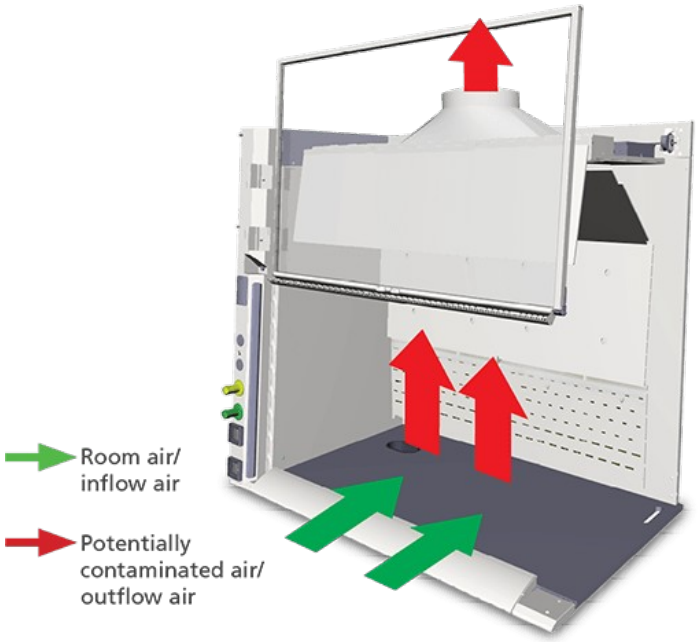
New Database with Updated Drop Table Queries (12/3/2025) Submit this one
https://drive.google.com/file/d/1GKN4x7f8I7gtu5BN6mzzMtcBuowhHJyK/view?usp=sharing

!!!! I found one small mistake in our database (in the CLIENT I wrote state C0 instead CO) That's why putting clients on the map didn't work correctly. So this database we should upload (12/5/2025).
https://drive.google.com/file/d/1iboWvzRanWjP2QCHA49If5EHl1UK1vjs/view?usp=sharing

Supplementary Materials



Picture 1. A scientist working in a fume hood. Source



Picture 2. Diagram of how air flows through a fume hood. Source

To Do Notes:

- Make sure we all can load in the DDL ~~(!!!! change the order of drop tables in the database)~~ ~~**Emily**~~ **done 12/3/2025**
- ~~Update tables names in ERD and schema for Order, Handle, and Uses~~
- ~~Make a couple changes to excel file and SQL DDL for the junction tables (Emily)~~
- Come up with questions we can ask of the database like assignment 3
    - Everyone come up with 2 questions to ask and write them down here
    - **Let's double check the queries!!!** Because I am not sure that it is 100% correct, plus maybe we can write the code more easily.
    - What about grouping out queries by categories for presentation?
        - According to the project details, we need to include 1 query per person, so we need 5 total. Maybe we can group them by category and then pick our favorite from there? That way we can demonstrate how the database is useful for a variety of purposes for the company - *Queries are categorized at the end of the doc*
- Make data visualization
- Once updates are made to ERD and schema, start working on presentation and add other stuff as its ready

Queries: (Queries sql file Queries for laboratory.sql)

-- 1. Find all fume hoods that have a "Fail" result in the most recent inspection table (included their department), and what materials/tests rely on them?

SELECT HoodID, DateRecentInspect, Result, TestID, LotNumber, Name AS MaterialName FROM INSPECT NATURAL LEFT JOIN FUMEHOOD
NATURAL LEFT JOIN USES NATURAL LEFT JOIN MATERIAL WHERE Result = 'Fail' ORDER BY HoodID, TestID;

-- 2. List all employees hired after 2023/01/01. Show EmployeeID, First, Last, HireDate, Salary, ordered by HireDate.

SELECT * FROM EMPLOYEE WHERE HireDate >= '2023/01/01' ORDER BY HireDate;

-- 3. Which employees handled the most tests (show EmployeeID, First, Last names, Number of handled tests)?

SELECT EmployeeID, First, Last, COUNT(TestID) AS Numb_Handled_Tests FROM EMPLOYEE NATURAL LEFT JOIN HANDLES GROUP BY EmployeeID, First, Last ORDER BY Numb_Handled_Tests DESC;

-- 4. Which employees work most often with which clients?

SELECT EmployeeID, First, Last, ClientID, CompanyName, COUNT(*) AS NumInteractions FROM HANDLES NATURAL LEFT JOIN EMPLOYEE
NATURAL LEFT JOIN CLIENT GROUP BY EmployeeID, First, Last, ClientID, CompanyName ORDER BY NumInteractions DESC, EmployeeID, ClientID;

-- 5. Which clients order the most tests?

SELECT ClientID, CompanyName, COUNT(TestID) AS Numb_Ordered_Tests FROM CLIENT NATURAL LEFT JOIN HANDLES GROUP BY ClientID, CompanyName ORDER BY Numb_Ordered_Tests DESC;

-- 6. Are some tests ordered by multiple clients (the most popular test)?

SELECT TestID, COUNT(ClientID) AS Numb_Ordered_Tests FROM HANDLES GROUP BY TestID ORDER BY Numb_Ordered_Tests DESC;

-- 7. Which types of materials are used in the most tests?

SELECT LotNumber, Name, COUNT(TestID) AS Numb_Used FROM MATERIAL NATURAL LEFT JOIN USES GROUP BY LotNumber, Name ORDER BY Numb_Used DESC;

-- 8. Which departments have the most fume hoods?

SELECT DepName, COUNT(HoodID) AS Numb_Fumehoods FROM DEPARTMENT NATURAL JOIN FUMEHOOD GROUP BY DepName ORDER BY Numb_Fumehoods DESC;

-- 9. Which materials expire in 2025 and 2026?

SELECT LotNumber, Name, ExpDate FROM MATERIAL WHERE YEAR(ExpDate) = 2025 OR YEAR(ExpDate) = 2026 ORDER BY ExpDate ASC;

-- 10. Which states generate the most orders?

SELECT State, COUNT(TestID) AS Numb_Orders FROM CLIENT NATURAL JOIN ORDERS GROUP BY State ORDER BY Numb_Orders DESC;

-- 11. Which department supports the most tests (via fume hoods)?

SELECT DepName, COUNT(*) AS TestsSupported FROM USES NATURAL LEFT JOIN FUMEHOOD NATURAL LEFT JOIN DEPARTMENT GROUP BY DepName ORDER BY TestsSupported DESC;

-- 12. Which fume hoods are used by the most tests?

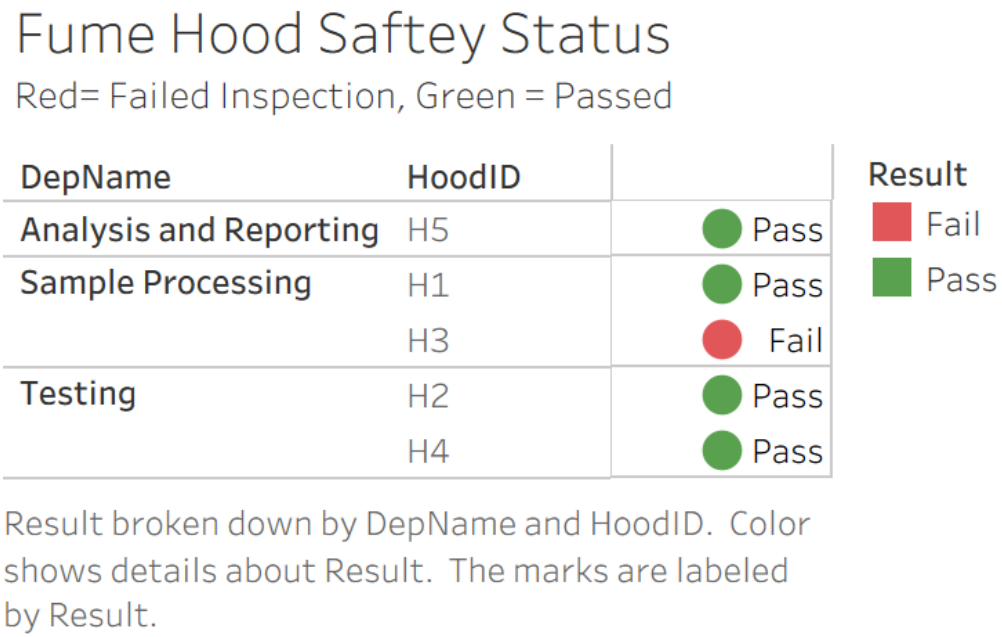
SELECT HoodID, COUNT(TestID) AS Numb_Tests FROM USES GROUP BY HoodID ORDER BY Numb_Tests DESC;

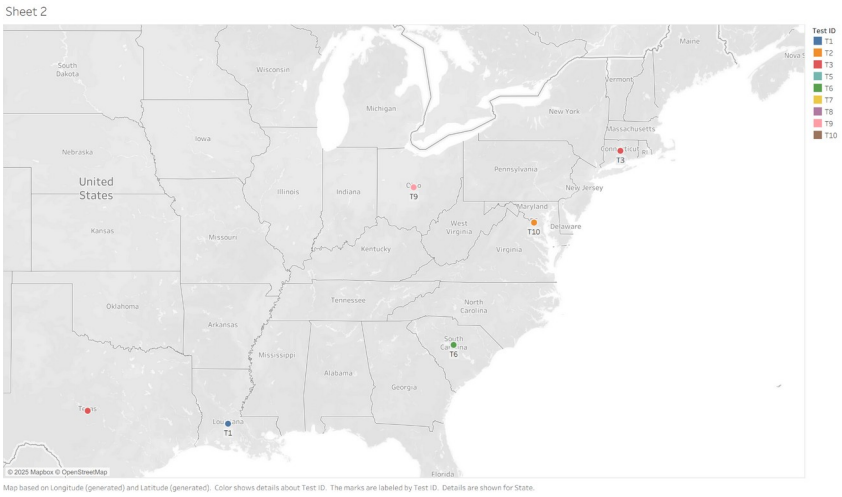
-- 13. Which clients' orders depend on failed fume hoods?

SELECT HoodID, TestID, ClientId, CompanyName FROM INSPECT NATURAL LEFT JOIN USES NATURAL LEFT JOIN ORDERS NATURAL LEFT JOIN CLIENT WHERE Result = 'Fail' GROUP BY CompanyName, ClientID, TestID;

Ideas for visualization:

1. U.S. Map: The number of clients by state
2. Timeline: Material Expiration Dates
3. Client Order Behavior Bar Chart (Top clients by number of tests)
4. Bar Chart: Most Frequently Ordered Tests





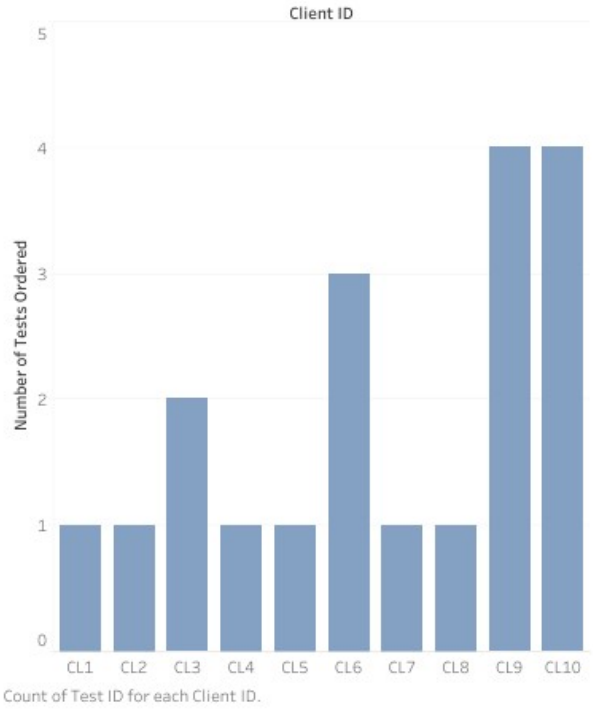
Another Idea for presentation

5405_Final_Project_Slides2.pptx

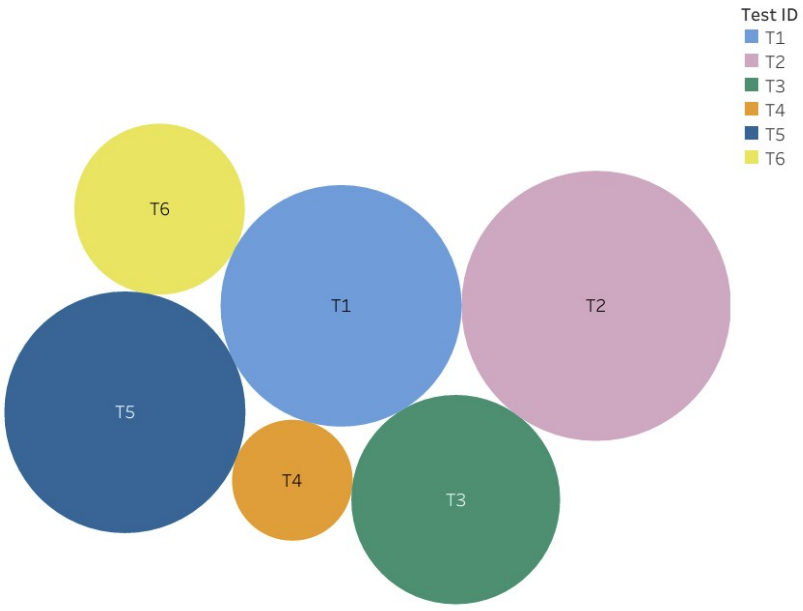
https://docs.google.com/presentation/d/1WuG8ZZlkKBfAxgzBop5MS6Za2jBv0QsHvvuWn8Q9NgQ/edit?usp=sharing

More Visualizations

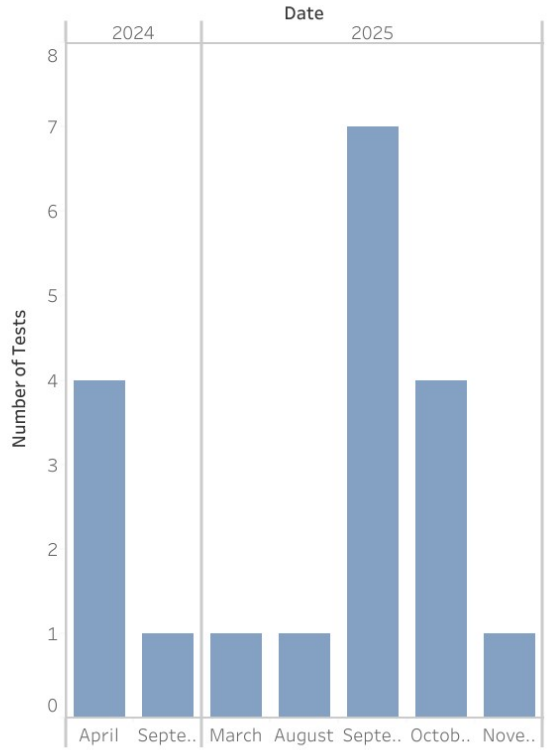
Tests Ordered by Clients



Count of Test ID for each Client ID.

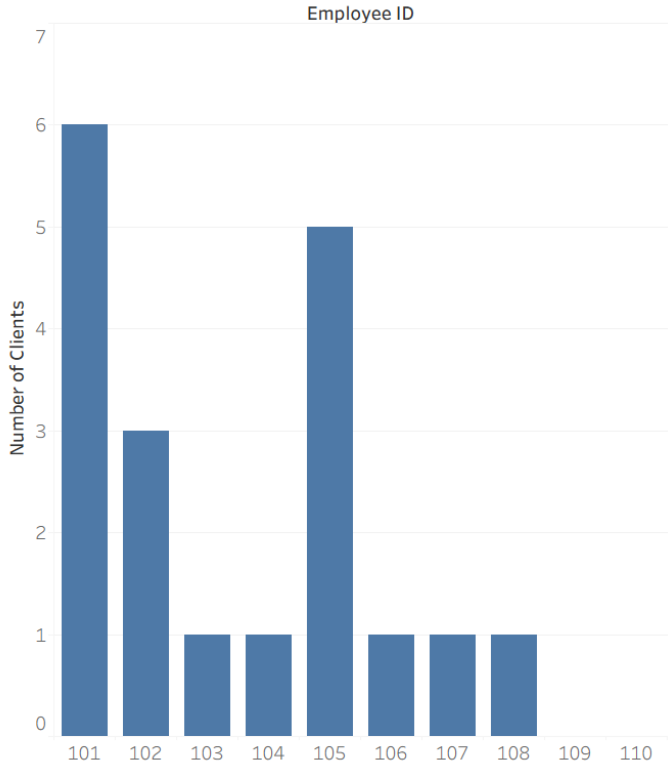


Tests Done per Month Each Year

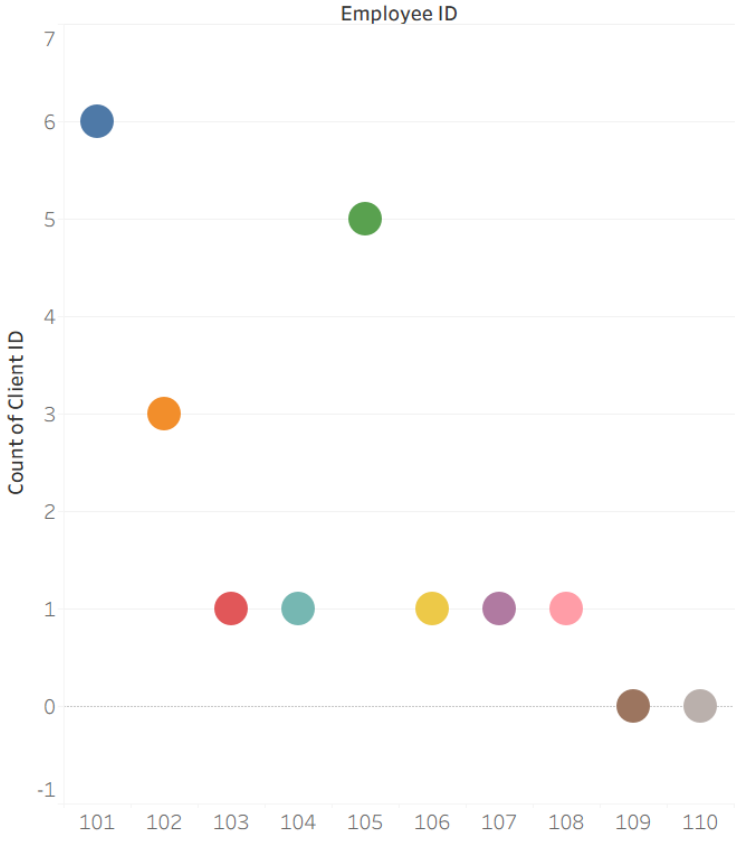


Count of Test ID for each Date Month broken down by Date Year.

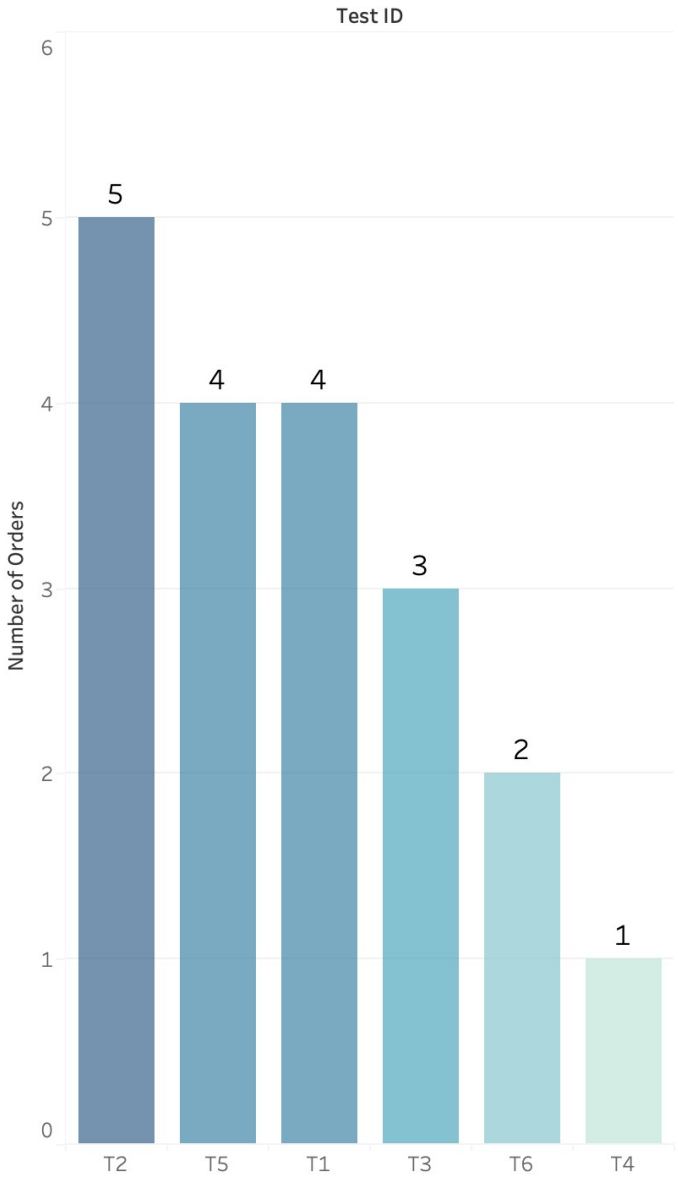
Number of Clients Worked with by Employee

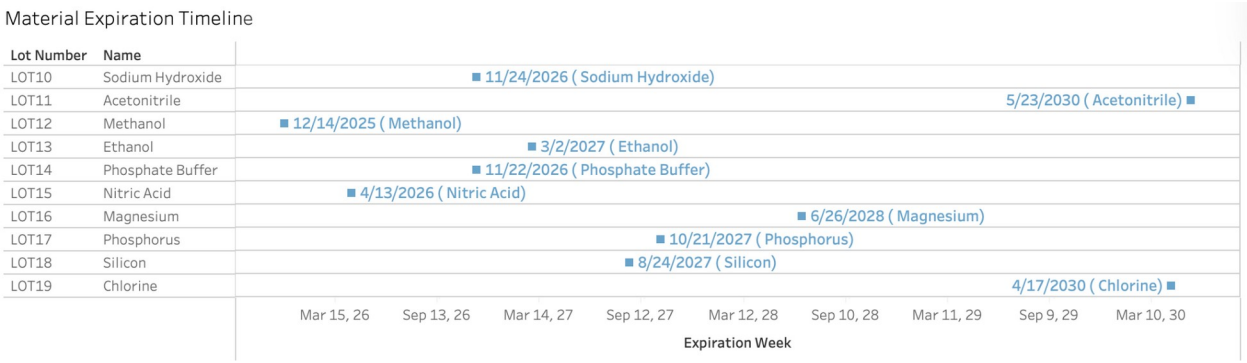


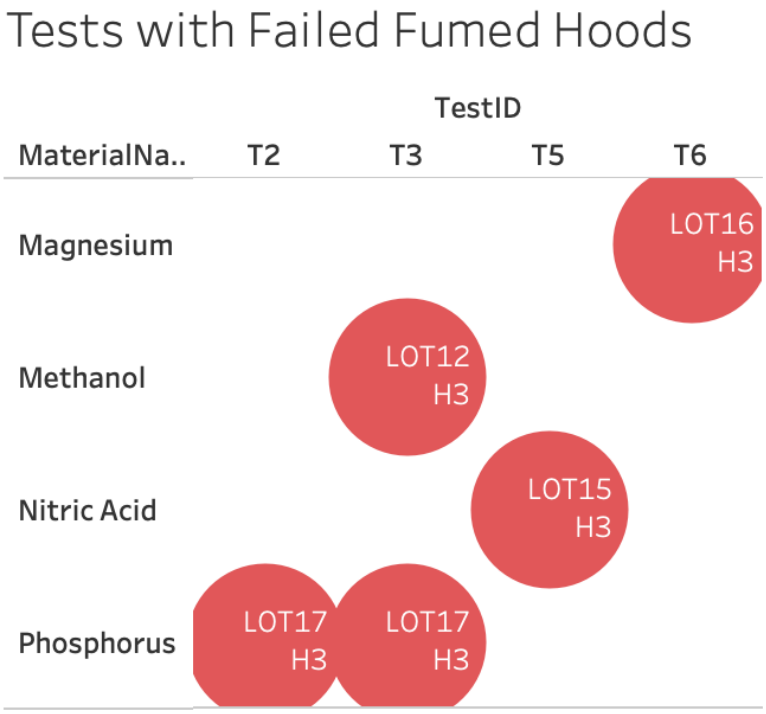
Number of Clients Worked with by Employee



Most Frequently Ordered Tests







Categorized Queries:

OPERATIONAL RISK & SAFETY:

1. Find all fume hoods that have a “Fail” result in the most recent inspection table (included their department), and what materials/tests rely on them? **Ryan**

SELECT HoodID, DateRecentInspect, Result, TestID, LotNumber, Name AS MaterialName FROM INSPECT NATURAL LEFT JOIN FUMEHOOD NATURAL LEFT JOIN USES NATURAL LEFT JOIN MATERIAL WHERE Result = 'Fail' ORDER BY HoodID, TestID;

-- 7. Which types of materials are used in the most tests?

SELECT LotNumber, Name, COUNT(TestID) AS Numb_Used FROM MATERIAL NATURAL LEFT JOIN USES GROUP BY LotNumber, Name ORDER BY Numb_Used DESC;

-- 8. Which departments have the most fume hoods?

SELECT DepName, COUNT(HoodID) AS Numb_Fumehoods FROM DEPARTMENT NATURAL JOIN FUMEHOOD GROUP BY DepName ORDER BY Numb_Fumehoods DESC;

-- 9. Which materials expire in 2025 and 2026?

SELECT LotNumber, Name, ExpDate FROM MATERIAL WHERE YEAR(ExpDate) = 2025 OR YEAR(ExpDate) = 2026 ORDER BY ExpDate ASC;

-- 11. Which department supports the most tests (via fume hoods)?

SELECT DepName, COUNT(*) AS TestsSupported FROM USES NATURAL LEFT JOIN FUMEHOOD NATURAL LEFT JOIN DEPARTMENT GROUP BY DepName ORDER BY TestsSupported DESC;

-- 12. Which fume hoods are used by the most tests?

SELECT HoodID, COUNT(TestID) AS Numb_Tests FROM USES GROUP BY HoodID ORDER BY Numb_Tests DESC;

-- 13. Which clients’ orders depend on failed fume hoods? **Anusuya**

SELECT HoodID, TestID, ClientId, CompanyName FROM INSPECT NATURAL LEFT JOIN USES NATURAL LEFT JOIN ORDERS NATURAL LEFT JOIN CLIENT WHERE Result = 'Fail' GROUP BY CompanyName, ClientID, TestID;

WORKFORCE & PRODUCTIVITY:

-- 2. List all employees hired after 2023/01/01. Show EmployeeID, First, Last, HireDate, Salary, ordered by HireDate.

SELECT * FROM EMPLOYEE WHERE HireDate >= '2023/01/01' ORDER BY HireDate;

-- 3. Which employees handled the most tests (show EmployeeID, First, Last names, Number of handled tests)? **Chirag**

SELECT EmployeeID, First, Last, COUNT(TestID) AS Numb_Handled_Tests FROM EMPLOYEE NATURAL LEFT JOIN HANDLES GROUP BY EmployeeID, First, Last ORDER BY Numb_Handled_Tests DESC;

-- 4. Which employees work most often with which clients?

SELECT EmployeeID, First, Last, ClientID, CompanyName, COUNT(*) AS NumInteractions FROM HANDLES NATURAL LEFT JOIN EMPLOYEE NATURAL LEFT JOIN CLIENT GROUP BY EmployeeID, First, Last, ClientID, CompanyName ORDER BY NumInteractions DESC, EmployeeID, ClientID;

REVENUE & CLIENT ANALYTICS:

-- 5. Which clients order the most tests? **Christina**

SELECT ClientID, CompanyName, COUNT(TestID) AS Numb_Ordered_Tests FROM CLIENT NATURAL LEFT JOIN HANDLES GROUP BY ClientID, CompanyName ORDER BY Numb_Ordered_Tests DESC;

-- 6. Are some tests ordered by multiple clients (the most popular test)? **Emily**

SELECT TestID, COUNT(ClientID) AS Numb_Ordered_Tests FROM HANDLES GROUP BY TestID ORDER BY Numb_Ordered_Tests DESC;

-- 10. Which states generate the most orders?

SELECT State, COUNT(TestID) AS Numb_Orders FROM CLIENT NATURAL JOIN ORDERS GROUP BY State ORDER BY Numb_Orders DESC;